

Haichao Zhu 朱海潮

Algorithm Engineer · Large Language Models & Search

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Profile

Algorithm engineer working on large language models, retrieval-augmented question answering, agents, search ranking, and text model alignment. Previously at Baidu's NLP and Search Strategy departments; selected for the AIDU program.

Experience

Text Model Alignment

July 2024 – Present

Algorithm Engineer · Beijing

Work on text model alignment.

Baidu · Natural Language Processing Department

June 2023 – June 2024

Algorithm Engineer · ERNIE Bot QA and Agents · Beijing

Worked on retrieval-augmented question answering, reasoning-driven retrieval, and general-purpose agent evaluation for ERNIE Bot.

Baidu · Search Strategy Department

July 2022 – June 2023

Algorithm Engineer · Search Strategy · Beijing

Selected for the Baidu AIDU program. Worked on end-to-end search ranking and content quality modeling.

Microsoft Research Asia

Oct. 2018 – Oct. 2019

Research Intern · Natural Language Computing Group · Beijing

Worked with [Dr. Furu Wei](#) and [Dr. Li Dong](#) on machine reading comprehension and text summarization with data augmentation.

Microsoft Software Technology Center Asia

May 2017 – Feb. 2018

Research Intern · Relevance and News Group · Beijing

Worked with [Dr. Furu Wei](#) and Yongliang (Ryan) Ma on multiple-choice machine reading comprehension.

Work Projects

ERNIE Bot: Retrieval-Augmented QA

June 2023 – March 2024

Built QA SFT data and improved retrieval content extraction, reranking, and answer freshness. In the reported Chinese QA evaluation, the GSB advantage over GPT-4 moved from -27.3% to +5.5%; extraction latency fell by 85%.

Reasoning-Driven Retrieval

October 2023 – February 2024

Designed retrieval triggers, query rewriting, knowledge organization, and integrated inference. Retrieval triggering decreased from 73% to 52%; the reported end-to-end GSB was 18:39:2 (+27 pp). Deployed as an ERNIE Bot plugin.

End-to-End Search Ranking

November 2022 – June 2023

Contributed quality and authority modeling in a multi-objective ranker. Overall PNR improved by 0.28, with quality +0.34 and authority +0.51. The system reduced 600+ features to 18, removed five manual strategies, and reduced 10+ ERNIE models to one.

End-to-End Quality Modeling

February 2023 – June 2023

Designed architecture, losses, inputs, features, and training for a multi-task quality model. The reported top-four high-quality result rate increased from 23.13% to 26.96%; offline PNR improved by 1.62.

General-Purpose Agent Evaluation

Built end-to-end and capability-level agent evaluation. Used automated feedback to guide data production and iteration; filtering low-scoring examples improved annotated-sample yield by 12%.

LLM Text Alignment

July 2024 – March 2025

Text model alignment.

Education

Ph.D. in Computer Science and Technology

September 2017 – July 2022

Harbin Institute of Technology · HIT-SCIR · Supervisor: Bing Qin

M.S. in Software Engineering

September 2015 – June 2017

Harbin Institute of Technology · HIT-SCIR · Supervisor: Bing Qin

B.E. in Computer Science and Technology

September 2010 – June 2014

Harbin Institute of Technology

Publications

SmartTrim: Adaptive Tokens and Attention Pruning for Efficient Vision-Language Models

Zekun Wang, Jingchang Chen, Wangchunshu Zhou, **Haichao Zhu**, Jiafeng Liang, Liping Shan, Ming Liu, Dongliang Xu, Qing Yang, Bing Qin

LREC-COLING · 2024

Learning to Tokenize for Generative Retrieval

Weiwei Sun, Lingyong Yan, Zheng Chen, Shuaiqiang Wang, **Haichao Zhu**, Pengjie Ren, Zhumin Chen, Dawei Yin, Maarten de Rijke, Zhaochun Ren

NeurIPS · 2023

TAGNet: A Tiny Answer-guided Network for Conversational Question Generation

Zekun Wang, **Haichao Zhu**, Ming Liu, Bing Qin

IJMLC · 2023

Distilled Dual-Encoder Model for Vision-Language Understanding

Zekun Wang, Wenhui Wang, **Haichao Zhu**, Ming Liu, Bing Qin, Furu Wei

EMNLP · 2022

Transforming Wikipedia into Augmented Data for Query-Focused Summarization

Haichao Zhu, Li Dong, Furu Wei, Bing Qin, Ting Liu

TASLP · 2022

Less Is More: Domain Adaptation with Lottery Ticket for Reading Comprehension

Haichao Zhu, Zekun Wang, Heng Zhang, Ming Liu, Sendong Zhao, Bing Qin

EMNLP Findings · 2021

Learning to Ask Unanswerable Questions for Machine Reading Comprehension

Haichao Zhu, Li Dong, Furu Wei, Wenhui Wang, Bing Qin, Ting Liu

ACL · 2019

Hierarchical Attention Flow for Multiple-Choice Reading Comprehension

Haichao Zhu, Furu Wei, Bing Qin, Ting Liu

AAAI · 2018

Skills

Python, C, C++, Shell · PyTorch, TensorFlow · Linux, Docker, Git